Detecting Language Associated With Home Healthcare Patient's Risk for Hospitalization and Emergency Department Visit

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Background: About one in five patients receiving home healthcare (HHC) services are hospitalized or visit an emergency department (ED) during a home care episode. Early identification of at-risk patients can prevent these negative outcomes. However, risk indicators, including language in clinical notes that indicate a concern about a patient, are often hidden in narrative documentation throughout their HHC episode.

Objective: The aim of the study was to develop an automated natural language processing (NLP) algorithm to identify concerning language indicative of HHC patients' risk of hospitalizations or ED visits.

Methods: This study used the Omaha System—a standardized nursing terminology that describes problems/signs/symptoms that can occur in the community setting. First, five HHC experts iteratively reviewed the Omaha System and identified *concerning concepts* indicative of HHC patients' risk of hospitalizations or ED visits. Next, we developed and tested an NLP algorithm to identify these concerning concepts in HHC clinical notes automatically. The resulting NLP algorithm was applied on a large subset of narrative notes (2.3 million notes) documented for 66,317 unique patients (n = 87,966 HHC episodes) admitted to one large HHC agency in the Northeast United States between 2015 and 2017.

Results: A total of 160 Omaha System signs/symptoms were identified as concerning concepts for hospitalizations or ED visits in HHC. These signs/symptoms belong to 31 of the 42 available Omaha System problems. Overall, the NLP algorithm showed good performance in identifying concerning concepts in clinical notes. More than 18% of clinical notes were detected as having at least one concerning concept, and more than 90% of HHC episodes included at least one Omaha System problem. The most frequently documented concerning concepts were pain, followed by issues related to neuromusculoskeletal function, circulation, mental health, and communicable/infectious conditions.

Conclusion: Our findings suggest that concerning problems or symptoms that could increase the risk of hospitalization or ED visit were frequently documented in narrative clinical notes. NLP can automatically extract information from narrative clinical notes to improve our understanding of care needs in HHC. Next steps are to evaluate which concerning concepts identified in clinical notes predict hospitalization or ED visit.

Key Words: clinical deterioration • home healthcare • natural language processing • nursing informatics • Omaha System • risk assessment

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Since the early 1970s, the healthcare system has drastically changed, with an increasing number of clinically complex patients treated in postacute care settings (e.g., long-term care hospitals, inpatient rehabilitation facilities, skilled nursing facilities, and home healthcare [HHC] agencies; Hardin & Mason, 2019; Jarvis, 2001). HHC has emerged as one of the many types of postacute care, represented by intermittent home visits conducted by skilled healthcare providers (e.g., registered nurses, physical therapists, and social workers). Currently, approximately 3.4 million adults receive HHC from more than 12,200 HHC agencies across the United States (Medicare Payment Advisory Commission, 2019). Previous studies support the critical role of HHC in demonstrating improved patient outcomes and reduced costs (Han et al., 2013; Howard et al., 2019; Weeks et al., 2018). HHC services are expected to increase because of longer life expectancy and a higher prevalence of chronic diseases among older adults (Landers et al., 2016; Mitzner et al., 2009).

Recent reports show that about one in five patients is hospitalized or visits an emergency department (ED) during HHC services (Medicare Payment Advisory Commission, 2019). Despite national and local quality improvement efforts focusing on acute and chronic ambulatory care-sensitive conditions (i.e., health conditions for which adequate management, treatment, and interventions delivered in the ambulatory care setting could potentially prevent hospitalization; Agency for Healthcare Research and Quality, 2018), these numbers have not improved over the last several years (Centers for Medicare & Medicaid Services, 2019). HHC providers have an opportunity to apply recent evidence, mainly from the hospital setting, to reduce hospitalization risk via early patient risk detection and notification (Fu et al., 2020). Given that up to 40% of hospitalizations are preventable with early or timely intervention (Morganti et al., 2013), the early identification of the patients at risk for clinical deterioration is emphasized as one of the key objectives to improve the quality of HHC.

The utilization of clinical nursing narrative documentation in predicting clinical outcomes is an area that has been coming into the limelight because notes often contain more information on patients' medical conditions than structured data in electronic health records or through standardized assessment tools (Song et al., 2021). In addition to standardized assessments, identifying risk factors in free-text data can further improve the prediction of healthcare utilization. Natural language processing (NLP) techniques used to extract the information in clinical notes can leverage free-text data more effectively and accurately. In the hospital settings, NLP was used to mine clinical notes written by nurses and other health providers to extract important symptom information (Koleck et al., 2021), identify sociobehavioral risk factors (e.g., substance or alcohol abuse; Navathe et al., 2018), find patients with depression (Zhou et al., 2015), predict patient's hospital mortality (Collins et al., 2013), and so forth. Several recent systematic reviews support the influence of NLP on a diverse range of medical fields, such as mental health (Le Glaz et al., 2021) and radiology (Sorin et al., 2020).

In HHC nursing, several studies demonstrated that nurses' expressions in free-text clinical notes were valid as an indicator for predicting negative outcomes such as deterioration (Topaz, Adams, et al., 2020; Topaz et al., 2021). However, those studies had a limited focus on a small number of risk factors (e.g., symptoms) when examining information from the clinical notes. Given that HHC is a community-based program providing services in patients' homes, inclusion of a larger set of personal, social, and environmental factors may improve identification of hospitalization or ED visit risk.

To address these knowledge gaps, this study aimed to develop a comprehensive NLP algorithm that can be used to create an early warning score system based on a set of lexicon categories for *concerning concepts* indicative of patient risk of hospitalizations or ED visits in HHC. Specifically, the aims were to (a) identify HHC concerning concepts that are indicative of an increased risk for hospitalization or ED visits using a standardized nursing terminology (the Omaha System), (b) create and validate an NLP algorithm to extract concerning concepts from narrative clinical notes, and (c) estimate the prevalence of concerning concepts in the clinical notes.

METHODS

Study Data Set and Population

The study data set included all patients admitted to HHC services at a large urban HHC organization in the Northeastern United States between January 1, 2015, and December 31, 2017. Patients could have more than one HHC episode; there were 87,966 HHC episodes for 66,317 unique patients. Episode was defined as all services provided within a specific time (i.e., initiated an admission and ended by a discharge). Approximately 2.3 million HHC clinical notes were extracted for this patient cohort. Two types of clinical notes were included:

1. visit notes, which are used to document the patient's status and the care provided during an HHC visit (total n = 1,029,535), and
2. care coordination notes, which are used to document communication between healthcare providers and to note other administrative care-related activities (total n = 1,292,442).

The institutional review boards approved the study of the participating institutions.

Identifying Concerning Concepts for Hospitalization or ED Use in HHC Patients

With 2.3 million notes to examine, the team used the Omaha System, a widely used standardized terminology for documentation of clinical information in community-based care (https://www.omahasystem.org/), to identify a subset of possible concerning terms (Martin, 2005). The Omaha System was chosen as the tool to examine documentation in this study because the system's standard terminology (i.e., ontology) comprehensively

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and holistically describes health factors to communicate and improve clinical practice (Monsen, 2018). The Omaha System includes 42 problems (e.g., income, neglect, healthcare supervision, and circulation), organized in four domains (environmental, psychosocial, physiological, and health-related behavior) and aligned with 335 unique signs/symptoms (e.g., uninsured medical expenses, lacks adequate physical care, inconsistent source of healthcare, and irregular heart rate).

In this study, we used the signs/symptoms and problem labels of the Omaha System to identify the corpus of terms used in NLP to detect concerning concepts documented in the nurse notes. The signs/symptoms and problem labels in the Omaha System associated with risk of hospitalizations or ED visits in HHC were defined as *concerning concepts*.

Concerning Concepts Concerning concepts were identified by five experts in HHC and informatics (J. S., M. T., K. H. B., A. T., and V. A.) based on the literature, expert's opinion, and/or their clinical experience. All had extensive clinical or research experience in HHC, one had a master's degree, and the other four had PhDs in nursing. The experts were presented with 335 Omaha System signs and symptoms and asked to indicate, "What signs/symptoms, if documented in the patient records, would cause concern for risk of hospitalizations or ED visits for HHC patients 65 years of age or older?"

The experts scored each sign/symptom on a scale from 1 to 3 (1 = *usually not concerning*, 2 = *occasionally concerning*, 3 = *usually concerning*). The signs/symptoms that received a score of 3 (*usually concerning*) from most experts (3 or more) were regarded as concerning signs/symptoms potentially associated with hospitalizations or ED visits in HHC. The study team discussed signs/symptoms that received a score of 2 from most experts for further consideration. Signs/symptoms that received a score of 1 from most experts (3 or more) were considered as "not concerning" (see the Supplemental Digital Content, http://links.lww.com/NRES/A431). The interrater agreement between experts was fair (Fleiss' κ = .32; McHugh, 2012). All discrepancies, including the sign/symptom with the difference between the maximum and the minimum score of 2, were resolved in several consensus group meetings.

NLP Algorithm Creation and Validation The process of the NLP algorithm creation and validation is depicted in Figure 1 and consists of six steps:

1. *Create a preliminary list of terms for concerning concepts using a vocabulary of standardized health terms.*

The initial steps to extract specific data require comprehensive lexicon development. We identified a list of synonyms for each concerning the Omaha System concept using a large vocabulary of standardized health terms (Unified Medical Language System [UMLS]; Sinha et al., 2013). For example, for the Omaha System sign/symptom of inadequate medication regimen under the problem of medication regimen, we identified UMLS synonym expressions such as "uses less medication than prescribed," "inconsistency with medication regimen," "no medication list," and "unsure of medication history."

Five of the research team members (J. S., M. O., M. T., V. A., and M. H.) reviewed the preliminary list independently and then validated and finalized the concerning signs/symptoms of the Omaha System. They were HHC nurses, PhD-level research scientists in nursing informatics, and a nursing PhD student.

2. *Language model creation: Word embedding model (Word2Vec).*

Language models are statistical representations of a certain body of text. A specific language model called word embedding (Word2Vec) was used (Mikolov et al., 2013). Word2Vec learns word associations from a large corpus of text; then once trained, it can help detect synonymous words for terms of interest based on a specific domain.

In this study, a publicly available NimbleMiner NLP software (Topaz et al., 2019) was used to identify synonyms for the concerning signs/symptoms present in a large body of HHC clinical notes available for the study period. The system can be downloaded from http://github.com/mtopaz/NimbleMiner under General Public License v3.0.

3. *Interactive vocabulary explorer to identify synonymous concerning signs/symptoms.*

An interactive rapid vocabulary explorer (part of the NimbleMiner software) was implemented by three coauthors (J. S., M. O., and M. T.), experts in HHC and informatics, to discover large vocabularies of synonyms. The researchers interacted with the software to input a query term of interest (e.g., sign/symptom of bruising under Omaha System problem of skin), and the system output included a list of potential synonyms (e.g., "bruise noted," "bruised area," and "ecchymosis"). We started the vocabulary expansion process using prepopulated lists of synonyms extracted from the UMLS for each concerning signs/symptoms in the Omaha System and then expanded the list of synonymous expressions using the interactive rapid vocabulary explorer. Relevant terms were selected and saved by clicking on them in the interactive vocabulary explorer user interface. The NLP software has a "negation module"; "negation" refers to words indicating negated synonyms (e.g., "denies," "no," "not," and "ruled out"). Hence, our NLP software can distinguish and exclude negated terms.

4. *Label assignment and review.*

In the final stage of NLP algorithm creation, the system used terms selected and saved during the previous stage to assign labels to clinical notes. Assigning a "positive label" means that the clinical note includes concerning signs/symptoms. The negation module distinguished the positive or negative sense of the concept of concern. For example, if a clinical note mentioned "severe dyspnea," it was considered a positive label. In contrast, clinical notes with a negation, such as "no dyspnea" or "denies dyspnea," were not assigned a positive label. We mapped the concerning signs/symptoms to a broader Omaha System problem category for labeling to enable

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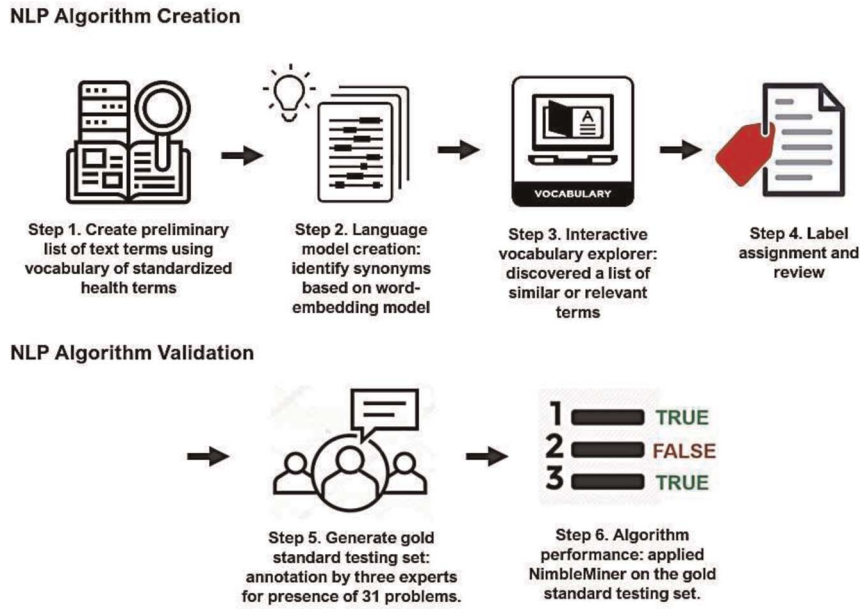


FIGURE 1. The process of workflow for natural language processing algorithm creation and validation.

human interpretability of the results and enable NLP system performance evaluation.

5. *Development of an expert-generated reference standard for NLP algorithm validation.*

To validate whether the NLP algorithm could identify Omaha System problems, including the concerning signs/symptoms with accuracy similar to a clinical expert, we generated a "gold-standard" testing set of 1,000 randomly extracted clinical notes. This gold-standard testing set was manually annotated using Microsoft Excel by three HHC experts (J. S., M. O., and M. T.) for the presence of concerning signs/symptoms at the level of the Omaha System problem. The observed interrater agreement was high (Fleiss' κ = .89), and all discrepancies were resolved through discussion.

6. *Evaluation of NLP algorithm performance.*

Lastly, the NLP algorithm was applied on the gold-standard testing set to calculate precision (defined as the number of true positives out of the total number of predicted positives), recall (the number of true positives out of the actual number of positives), and *F* score (the weighted harmonic means of the precision and recall) for each Omaha System problem.

Applying the NLP Algorithm The finalized NLP algorithm was applied to the entire data set, including approximately 2.3 million HHC clinical notes. Descriptive statistics were used to calculate the proportion of notes that include concerning signs/symptoms. The differences in note length and type of note between the notes that had the concerning signs/symptoms and those that did not include the concerning signs/symptoms were compared using Student's *t* test or chi-square tests. For all analyses, *p* values less than .05 were regarded as the indicator of statistical significance. Odds ratios for the type of notes were calculated with 95% confidence intervals. Lastly, the frequency of each Omaha System problem was calculated at the episode level. All analyses were implemented using R software Version 4.1.0 (R Foundation of Statistical Computing, Vienna, Austria).

RESULTS

Identifying Concerning Signs and Symptoms for Hospitalizations and ED Visits in HHC

Of 335 signs/symptoms of the Omaha System presented to the experts, 131 were initially identified as *concerning*, with 29 signs/symptoms added after reviewing the "occasionally concerning" category through group consensus. This resulted in a list of 160 signs/symptoms identified as concerning signs/symptoms for hospitalizations or ED visits in HHC. These signs/symptoms belong to 31/42 (73.8%) of available Omaha System problems. In several Omaha System problem categories (e.g., abuse), virtually all signs/symptoms were considered concerning signs/symptoms. In other problem categories, only selected subsets of signs/symptoms were selected (e.g., circulation, skin, or income). For instance, under the skin problem, the signs/symptoms of excessively dry, excessively oily, or hypertrophy of nails were not considered concerning concepts for the risk of unplanned hospitalization or ED visits. Furthermore, no signs/symptoms from several problem categories (e.g., postpartum or pregnancy) were selected concerning signs/symptoms because they did not apply to the older adult HHC patient population of interest for the study. The detailed concerning signs/symptoms of the Omaha System problem is

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presented in the Supplemental Digital Content (http://links.lww.com/NRES/A431).

NLP Algorithm Performance

Among 1,000 clinical notes randomly selected for gold-standard manual review, 393 (39.3%) included at least one Omaha System problem, and a total of 496 Omaha System problems were detected. The most common Omaha System problem documented in the clinical notes was neuromusculoskeletal function (16%), which included signs/symptoms of decreased balance and gait/ambulation disturbance, followed by pain and circulation problems as the next most common documented problems (14% and 9%, respectively). The overall NLP algorithm's performance in identifying the Omaha System problem, including the concerning signs/symptoms, was good (average *F* score = 0.84), with best results for the social contact, hearing, income, and abuse problems (*F* score = 1) and poorest results for the nutrition problem (*F* score = 0.62; see Table 1).

Applying NLP Algorithm on a Large Set of Clinical Notes

On average, 26.4 (*SD* = 27.8) clinical notes were documented for each episode. In general, visit notes were longer than care coordination notes (mean length [*SD*] = 246.8 [241.4] characters vs. 99 [98.5] characters, $p < .05$).

When applying the NLP algorithm on the full sample of clinical notes, 18.1% (419,517 of 2,321,977) notes were detected as having at least one concerning sign/symptom. Among notes that included at least one concerning sign/symptom, 81% (347,652 of 419,517) visited notes, and the rest were care coordination notes. Statistically, concerning signs/symptoms were almost nine times more likely to be documented in visit notes than in care coordination notes (*OR* = 8.7, 95% CI [8.6, 8.8]). In addition, the average length of notes that included concerning signs/symptoms was significantly longer than notes that did not have concerning signs/symptoms (mean length [*SD*] = 359.4 characters [263.6] vs. 121.6 characters [138.4], all *p*s < .05). Below are example sentences that include concerning signs/symptoms:

Example 1: "…VN (visiting nurse) reported to MD (medical doctor) *swelling (concerning signs/symptoms under the problem of circulation) and bruise (concerning signs/symptoms under the problem of skin) noted on pts (patient) lt (left) lower leg and foot*."

Example 2: "…PT (patient) is homebound due to *weakness on both legs (concerning signs/symptoms under the problem of neuromusculoskeletal function) and difficulty walking (concerning signs/symptoms under the problem of neuromusculoskeletal function)*."

Example 3: "…MD (medical doctor) saw patient earlier in week and patient had congestion. Reviewed patient's diet: *patient is not compliant with low salt diet (concerning signs/symptoms under the problem of healthcare supervision)*."

A total of 289,324 Omaha System problems were detected during the study period. Figure 2 shows the frequency of episodes with documented Omaha System problems in the clinical notes. At the episode level, the most frequently reported problem was pain, followed by neuromusculoskeletal function, circulation, mental health, and communicable/infectious conditions (48.3%, 46.1%, 35.3%, 32.8%, and 25.9%, respectively). Overall, an average of 3.2 (*SD* = 2.3, median = 3) Omaha System problems were detected per episode. Among the episodes having at least one Omaha System problem, an average of 3.6 (*SD* = 2.2, median = 3) problems were detected. Over 90% of the home health episodes had at least one Omaha System problem (i.e., with concerning signs/symptoms) documented (79,865 out of 87,966).

DISCUSSION

To our knowledge, this is the first study to identify concerning concepts potentially associated with hospitalization or ED visits in HHC using the Omaha System. Compared to other standardized terminologies such as the International Classification of Diseases, Ninth Revision, Clinical Modification that encompasses mostly medical diagnoses, the Omaha System was developed to describe the healthcare provided in community settings, like HHC and public health. Previously, the Omaha System has been identified as a better candidate for predicting hospitalization in HHC patients compared to the International Classification of Diseases, Ninth Revision, Clinical Modification (Monsen et al., 2012).

Our results highlight the uniqueness of the HHC environment. Compared to risk factors identified in a similar study in the hospital setting (Kang et al., 2020), we found additional unique signs/symptoms under the healthcare supervision problem (e.g., fails to obtain routine/preventive care or inconsistent source of healthcare) and the medication regimen problem (e.g., not follow recommended dosage/schedule or unable to take medications without help). A recent systematic review showed that most of the previous studies of HHC patient risk of hospitalization and ED visit only used a small fraction of potential risk factors at the level of signs/symptoms of the Omaha System identified in this study (Ma et al., 2018). For example, concerning concepts such as signs/symptoms within healthcare supervision or medication regimen of the Omaha System, problems were infrequently, if ever, identified in previous studies. To develop a more comprehensive picture for modeling HHC patient risk, more inclusive terms, such as those in the Omaha System, are warranted.

Our results are supported by a recent study using a different form of NLP to identify risk factors associated with HHC patient risk for hospitalizations and ED visits (Topaz, Woo, et al., 2020). This previous study found several broad categories of words and expressions HHC nurses use to

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TABLE 1. Evaluation of Natural Language Processing Algorithm Performance Via Gold-Standard Manual Review (Total *N* = 1,000 Clinical Notes)

The Omaha System problems	Total frequency and proportion of documentation, % (*n*)	Precision	Recall	*F* score
Neuromusculoskeletal function	16 (78)	0.99	0.76	0.86
Pain	14 (68)	0.84	0.95	0.89
Circulation	9 (47)	0.94	0.80	0.86
Mental health	9 (47)	0.96	0.75	0.84
Skin	9 (46)	0.93	0.80	0.86
Healthcare supervision	7 (35)	1.00	0.61	0.76
Cognition	7 (34)	0.97	0.89	0.93
Respiration	6 (32)	0.97	0.78	0.86
Communicable infectious condition	4 (18)	0.94	0.81	0.87
Social contact	3 (17)	1.00	1.00	1.00
Digestion hydration	3 (14)	0.93	0.62	0.74
Medication regimen	2 (9)	0.88	0.68	0.77
Bowel function	2 (8)	1.00	0.89	0.94
Genitourinary function	2 (8)	1.00	0.73	0.84
Nutrition	2 (8)	1.00	0.44	0.62
Neglect	1 (5)	1.00	0.45	0.63
Communication with community resources	1 (4)	1.00	0.57	0.73
Speech and language	1 (4)	1.00	0.8	0.89
Hearing	1 (3)	1.00	1.00	1.00
Income	1 (3)	1.00	1.00	1.00
Residence	1 (3)	1.00	0.70	0.82
Consciousness	0 (2)	1.00	0.71	0.83
Abuse	0 (1)	1.00	1.00	1.00
Interpersonal relationship	0 (1)	0.50	1.00	0.67
Personal care	0 (1)	1.00	0.67	0.80
Total	100 (496)			
Average		0.95	0.78	0.84

Note. Only categories where *concerning concepts* found in gold-standard references were presented. Therefore, neighborhood/workplace safety, oral health, sanitation, sexuality, sleep and rest patterns, and substance use were not presented in the table.

indicate increased patient's risk, including "clinical factors" (expressions like "nausea," "dehydrated," and "vomited"), "coordination/communication" (expressions like "plan discussed," "MD called," and "supplies ordered"), and "social/environmental factors" (expressions like "lives alone," "PT lives with," and "private insurance"). Virtually, all of these broad categories are represented by the Omaha System problems identified in this study. Our results represent a more complete list of risk factors than the previous research.

This study aimed to develop and test an NLP algorithm to identify factors related to the risk of hospitalization or ED visits in HHC. The NLP algorithm achieved good performance on a subset of randomly selected HHC clinical notes. Our findings suggest that the risk of hospitalization or ED visits was frequently documented in narrative clinical notes. NLP can extract data from narrative clinical notes to improve our understanding of care needs in HHC. This finding also highlights the value of HHC clinical documentation in improving identification of patients at risk for hospitalization or ED visits (Song et al., 2021; Topaz, Woo, et al., 2020).

In addition, in developing the NLP algorithms, our team used clinician-validated lexicons rather than unsupervised machine learning techniques. Although sometimes machine learning techniques can capture hidden patterns that are hard to identify through human review (Wang et al., 2019), the lexicons created by clinical experts are transparent, comprehensible, and intuitive (Weng et al., 2017). Our study adds to the research conducted by others that successfully used the Omaha System in text mining applications (Bjarnadottir et al., 2020; Monsen et al., 2018). These results can inform the potential development of an early warning scoring system to help reduce negative outcomes in HHC by identifying a patient's concerns, signs, and symptoms.

In this study, 90% of HHC episodes had at least one documented Omaha System problem. This number is higher than reported in previous NLP studies. For example, recent studies examined HHC clinical notes for presence of six common symptoms (i.e., pain, fatigue, sleeping difficulty, breathing difficulty, depressed mood, and anxiety) and found that two thirds of HHC episodes had one of these symptoms

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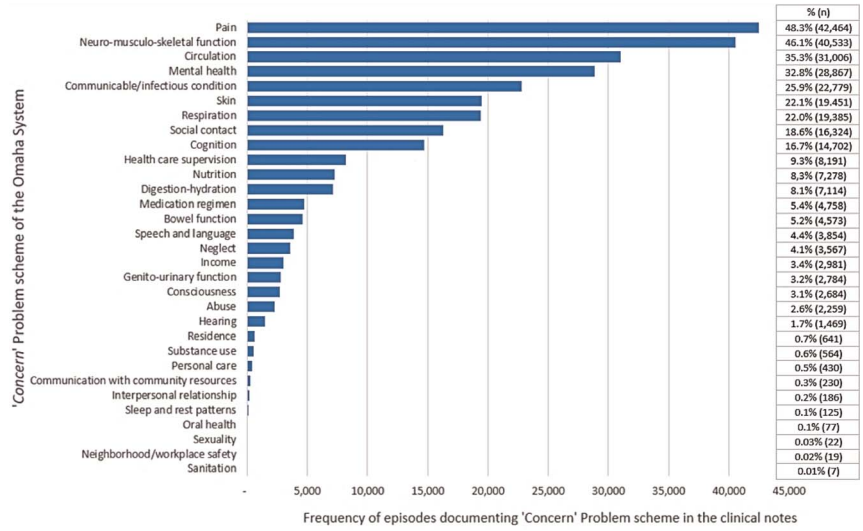


FIGURE 2. Frequency of home healthcare episodes containing documentation of concerning signs and symptoms within Omaha System problems.

(Patel et al., 2019; Topaz et al., 2021). As the Omaha System implemented in this study included the above six symptom categories and many additional problems, our study expands the previous findings. The presence of problems in many HHC episodes is also an expected clinical finding; to be eligible for HHC services, patients need to present one or more problems that require intervention by skilled HHC providers. Concerning signs/symptoms were more frequently documented in visit notes than in care coordination notes. This is an expected difference; visit notes often include information about the patient's signs and symptoms (the focus of this study), whereas care coordination notes often describe issues related to care management, communication, and care plans.

Among the Omaha System problems, pain was most frequently documented, followed by neuromusculoskeletal function, circulation, mental health, and communicable/infectious conditions. The prevalence of problems noted in our study was similar to that of other studies. For example, other studies report that one half of a community-dwelling elderly population experienced acute/chronic pain (Hunt et al., 2015; Patel et al., 2019); our findings show a similar percentage (48.3%). Similarly, our results suggest that mental health issues, including fatigue, anxiety, and depressed mood, were documented for over one third of HHC episodes. This is consistent with the rate in the community-dwelling older adults who suffered from those issues, as reported in other studies (de Rekeneire et al., 2014; Zengarini et al., 2015). On the other hand, problems of sexuality, neighborhood/workplace safety, and sanitation were rarely documented. This study was limited to one agency; perhaps HHC clinicians in other geographic locations would encounter different concerns. This finding needs to be explored further using qualitative and quantitative methods.

Our findings can inform several further studies. First, we plan to conduct a large retrospective study that will examine the association between the documentation of risk factors—including the type of concerning signs/symptoms, along with the combination of and intensity of the documented concerning signs/symptoms—and health outcomes (i.e., hospitalization or ED visits in HHC). Our early results in other studies support strong associations between documented symptoms and patient outcomes in HHC (Topaz, Adams, et al., 2020; Topaz et al., 2021; Wang et al., 2019). We expect to find similar correlations with concerning signs/symptoms. Our NLP system can also be used as a foundation for further development of an early warning system. Specifically, the suggested early warning system will automatically scan HHC clinical notes to identify concerning signs/symptoms. Once concerning signs/symptoms are identified, they will be communicated in automated electronic alerts with the HHC nurse who is managing the patient and potentially HHC clinical managers or primary care providers when necessary. As shown in hospital settings (Lee et al., 2020), alerts generated by such early warning systems can help provide timely interventions to reduce patients' risk for negative outcomes.

Lastly, recent studies (Moy et al., 2021), including in HHC (Hobensack et al., 2021), show that clinicians experience high documentation burden with up to 50% of clinical time spent on documentation. Some documentation is redundant as some concerning signs/symptoms are documented in a standardized form (e.g., drop-down list of several types of patient's breathing issues) and in clinical notes. Our results show that HHC clinical notes are a rich source for concerning signs/symptoms. Hypothetically, some redundant standardized form documentation can be removed or shortened to reduce clinical documentation burden. However, further research is needed to understand the completeness of different documentation

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types (e.g., standardized vs. clinical notes) and to examine the amount of time it takes to document using one type of documentation versus the other.

Limitations

This study has several limitations. First, we identified the signs/symptoms of the Omaha System indicative of the risk for hospitalization and ED visits in HHC through expert reviews. Therefore, additional external validation is required to integrate the classification into a real-world clinical decision support system (e.g., early warning score system). Second, as this study was conducted using data from a single organization, there is a possibility that organization-specific clinical language and jargon patterns limit the generalizability of the NLP algorithms. Third, the data were collected in 2015–2017, which may not appropriately reflect documentation trends in more recent data.

Conclusion

This study is the first to develop and test an NLP algorithm to identify concerning concepts related to the risk of hospitalization or ED visits in HHC using standardized health terminology (Omaha System). Our findings suggest that information concerning signs and symptoms that may indicate the risk of hospitalization or ED visits was frequently documented in narrative clinical notes. NLP can extract information from narrative clinical notes to improve our understanding of care needs in HHC. The study findings provide a foundation for developing an early warning scoring system to help reduce adverse outcomes in HHC.

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The Columbia University and Visiting Nurse Service of New York institutional review boards approved this study.

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